CSA1701-ARTIFICIAL INTELLIGENCE LAB EXPERIMENTS

Program 8 : Python Program to Implement DFS

Aim

To develop a Python program that traverses a graph using the Depth First Search (DFS) algorithm.

Algorithm

- Represent the graph as an adjacency list using a dictionary.
- Read the starting node and call the recursive function dfs().
- If the node is not visited, print it and add it to the visited list.
- For each neighbour of the node, call dfs() recursively.
- Stop when all reachable nodes have been visited.

Program

```
graph = {     'A':
['B', 'C'],
    'B': ['D', 'E'],
    'C': ['F'],
    'D': [],
    'E': [],
    'F': []
} visited =
[]

def dfs(node):     if node
not in visited:
print(node, end=" ")
visited.append(node)

        for next_node in graph[node]:
dfs(next_node)

start = input("Enter starting node: ")
dfs(start)
```

Output

```
Python 3.14.3 (tags/v3.14.3:323c59a, Feb  3 2026, 16:04:56) [MSC v.1944 64 bit (
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
============== RESTART: C:/Users/shaik/OneDrive/Documents/8.py ==============
Enter starting node: A
A B D E C F
>>>
```

RESULT

Thus, the DFS traversal was implemented successfully using Python.